



INTRO TO CAD

UNIT 2: DESIGNING A **SKATEBOARD**

Lesson 1 Student Guide

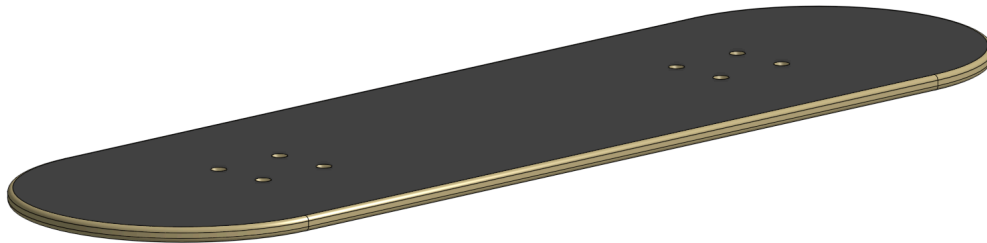
Lesson 2.1 Designing the Deck

Student Guide

Designing a Skateboard

We are going to start designing a skateboard today! Have you skateboarded before? Do you own a skateboard? Skateboards are a great place to jump into making your own models in CAD. We will guide you through the steps to make a skateboard and you'll even get a chance to attach parts and make them spin!

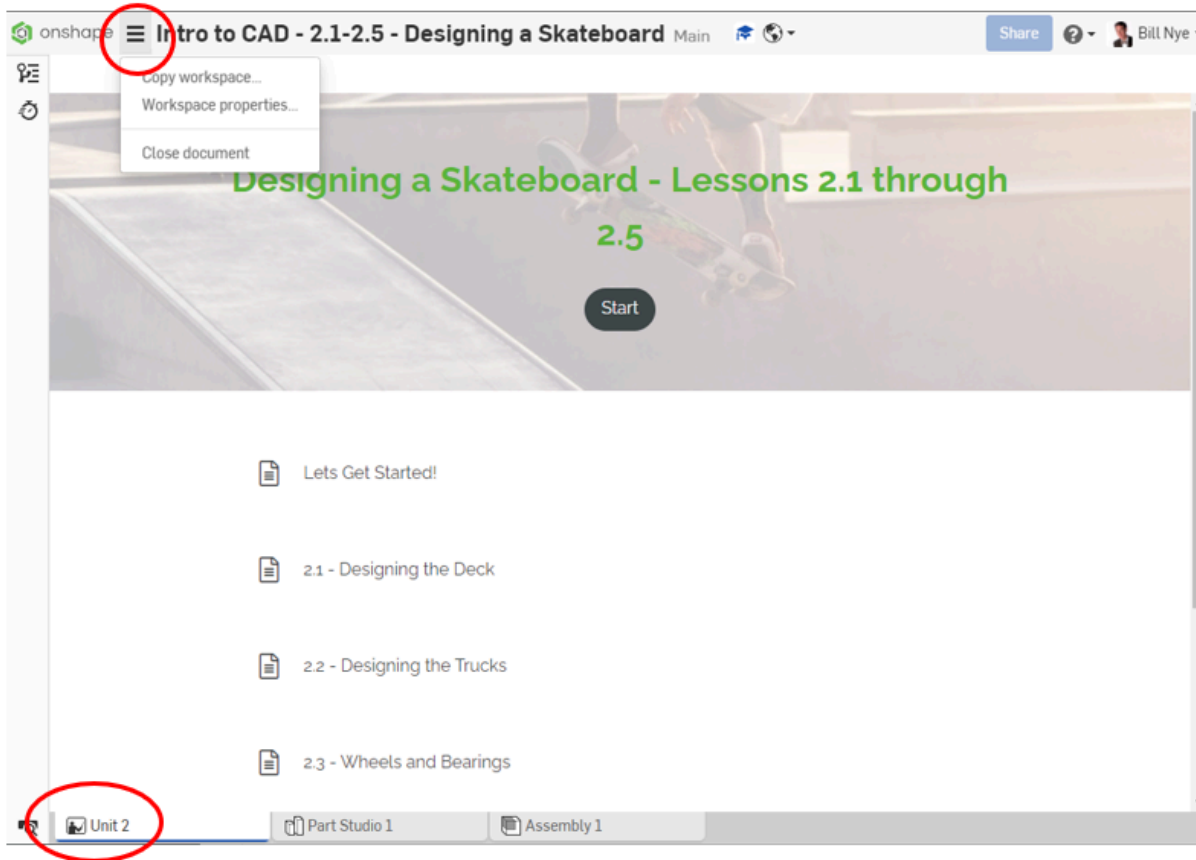
During these lessons you'll follow pretty specific instructions on how to build the skateboard. However, you will get to modify this board and make it your own by the end of this unit. So, hang on!



Lesson 2.1: Designing the Deck

1. Open the document [Intro to CAD - 2.1-2.5 - Designing a Skateboard](#)
2. Click the Document menu button in the upper left.
3. Select "Copy workspace..."
4. Rename the document to "Intro to CAD - 2.1-2.5 - [Your Name]".
5. Open **Unit 2** tab within the document.
6. Begin!

Document menu button



Assignment Criteria

Your deck sketch is fully constrained.	4 pts	
Exit Ticket #1	3 pts	
Exit Ticket #2	3 pts	
Total	10 pts	/10

Your Onshape Document Link

Paste your document URL here:

<https://cad.onshape.com/documents/1635f53ab68046523fefaa5d/w/169387cdbdd53f7d08136dc2/e/e64bb8de2d3ef829f45b1dgo>

Exit Ticket

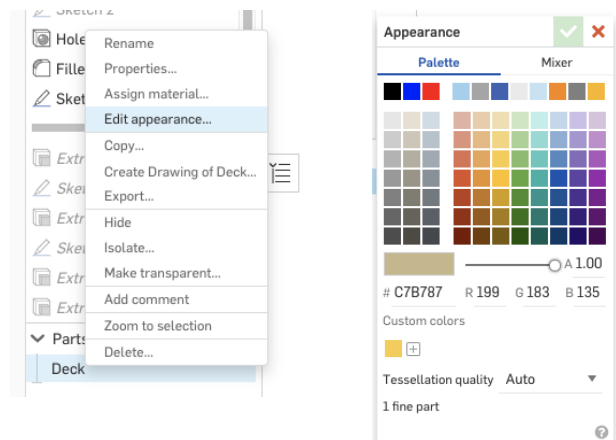
- 1) Make sure you have applied the material "Hard Maple (Density 0.026 lb/in³)". What is the mass of the skateboard deck? 3.488
- 2) If the skateboard deck changed size (width and/or length), would the bolt holes remain centered on the board? Why or why not? They would not because it won't change the material of it just the size

Extension

Got your skateboard deck done early? Modify your skateboard to make it your own! Here are some things to try.

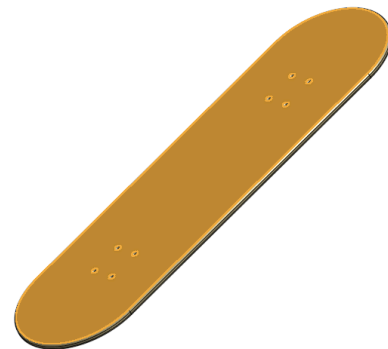
Change the appearance of the deck.

- Right click on the Deck
- Select **Edit appearance**
- Choose a color



Change the surface

- Add some cool grip tape!
- Select just the surface of the board
- Right click the surface
- Select **Edit face appearance**
- Choose a color



Looking for more? Add a design to the bottom of the board.

- Start a new **Sketch** on the bottom surface of the board
- Play with the sketch tools to create a design of your own
- Extrude your design .125 in